Akoizo Original Practice - Form 1

Division C | Ping Pong Parachute | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Three timers read 12.41, 12.56, and 12.30 s. What is the official time?

A. 12.41 s

B. 12.56 s

C. 12.42 s

D. 12.30 s

Answer: _____

2. (2 points) Which deployment method is allowed?

Unit: Deployment and separation.
Worked-case data: Video shows a payload leaving the rocket 0.3 s after launch, well before apogee, and the parachute tangling.
What should change?
Separate investigation observations:
Cup depth 2 cm: Payload left during climb in 4 of 5 flights.
Cup depth 4 cm: Payload stayed attached in 3 of 5 flights.
Cup depth 3 cm: Payload released near apogee in 5 of 5 flights.

A. A small electronic timer

B. A rubber band ejector

C. A loose nose cup that releases as the rocket slows at apogee

D. A spring that pushes the parachute out

Answer: _____

3. (2 points) Why does the pressure-height curve flatten at high pressure?

Unit: Apogee, ceiling height, and choosing pressure.
Worked-case data: Your chart shows 50 psi -> 17 ft, 60 psi -> 21 ft, 70 psi -> 24 ft, with about ±1.5 ft spread. The ceiling is 25 ft. Which pressure should you choose?
Separate investigation observations:
Chart: 45 psi -> 15 ft, 55 psi -> 19 ft, 65 psi -> 22 ft.
Spread: About ±1 ft at each pressure.
Venue: Ceiling announced at 21 ft.

A. The bottle holds less air

B. Timers stop earlier

C. Drag grows rapidly with speed

D. Gravity gets stronger with height

Answer: _____

4. (2 points) Which phase of the flight earns most of the time aloft?

Unit: How an air-powered bottle rocket climbs.
Worked-case data: A rocket reaches apogee 1.2 s after launch, and the payload then descends for 11.5 s. What share of time aloft came from the descent?
Separate investigation observations:
40 psi: Peak about 12 ft; time aloft 8.1 s.
55 psi: Peak about 18 ft; time aloft 12.4 s.
70 psi: Peak about 24 ft; time aloft 16.0 s in a 30 ft hall.

A. The coasting climb

B. The time on the launcher

C. The parachute descent

D. The powered phase

Answer: _____

5. (2 points) Time aloft plotted against peak height forms a straight line. What does the slope tell you?

A. Seconds of descent gained per foot of height

B. The launch pressure

C. The ceiling height

D. The rocket’s mass

Answer: _____

6. (4 points) Explain why a parachute reaches a steady falling speed.

Unit: Parachute physics: drag and slow descent.
Worked-case data: A 4.7 g payload uses a 40 cm diameter dome (C_d about 1.5). Estimate its descent speed and the time to fall 7 m.
Separate investigation observations:
30 cm canopy: Opens every time; 7.8 s descent.
60 cm canopy: Opens 6 of 10 times; 14.9 s when open, 3 s when not.
50 cm canopy, looser fold: Opens 10 of 10; 12.6 s descent.

Answer: _____
_____
_____

7. (4 points) Describe how to check that a nose meets the rules.

Unit: Building a legal, stable rocket.
Worked-case data: A rocket's CG is 18 cm from the nose and its CP is 21 cm from the nose. The bottle diameter is 8 cm. Is it stable enough?
Separate investigation observations:
Flight 1: Rocket corkscrews and lands near a wall.
Inspection: One fin is twisted about 8° from the body line.
Flight 2: Fin realigned: rocket flies straight up.

Answer: _____
_____
_____

8. (4 points) What two consequences follow from not having a practice log?

Unit: The event and its score: time aloft and multipliers.
Worked-case data: A team with a complete log records 14.20 s on launch 1 (clean) and 19.80 s on launch 2, but launch 2 grazed a light on the way up. What is its score?
Separate investigation observations:
Ceiling: 24 ft gym.
Log data: 60 psi: about 20 ft peak, 15 s aloft; 75 psi: about 26 ft peak.
Team plan: Launch at 75 psi for "more height."

Answer: _____
_____
_____

9. (4 points) What happens if the payload releases too early?

Unit: Deployment and separation.
Worked-case data: Video shows a payload leaving the rocket 0.3 s after launch, well before apogee, and the parachute tangling. What should change?
Separate investigation observations:
Cup depth 2 cm: Payload left during climb in 4 of 5 flights.
Cup depth 4 cm: Payload stayed attached in 3 of 5 flights.
Cup depth 3 cm: Payload released near apogee in 5 of 5 flights.

Answer: _____
_____
_____

10. (4 points) Give one source of error in angle measurements and one way to reduce it.

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: A rocket reaches apogee 1.2 s after launch, and the payload then descends for 11.5 s. What share of time aloft came from the descent?

Answer: _____
_____
_____

12. (6 points) Solve this case and explain your reasoning: A log has 12 flights recording pressure, height, time, and fin count. Is it complete?

Answer: _____
_____
_____

13. (2 points) Which parachute should the team choose?

Supplied synthetic observations:
30 cm canopy: Opens every time; 7.8 s descent.
60 cm canopy: Opens 6 of 10 times; 14.9 s when open, 3 s when not.
50 cm canopy, looser fold: Opens 10 of 10; 12.6 s descent.

A. The 50 cm canopy with the looser fold

B. The 60 cm canopy

C. The 30 cm canopy

Answer: _____

14. (2 points) Define “Center of gravity (CG)” in the context of Ping Pong Parachute.

Answer: _____
_____
_____

15. (2 points) Define “Parachute payload system” in the context of Ping Pong Parachute.

Answer: _____
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16. (2 points) Define “Passive deployment” in the context of Ping Pong Parachute.

Answer: _____
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17. (2 points) Define “Canopy area” in the context of Ping Pong Parachute.

Answer: _____
_____
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18. (2 points) Define “Cap test” in the context of Ping Pong Parachute.

Answer: _____
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